



IILM
UNIVERSITY

IILM University, Greater Noida
SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Healthcare Analytics System

A Database Management System (DBMS) Project

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ABSTRACT

The Healthcare Analytics System is a database management system designed to efficiently store, manage, and analyze healthcare-related data. The system maintains records of patients, doctors, appointments, diagnoses, treatments, and billing information in a structured relational format.

The objective of this system is to reduce redundancy, ensure data integrity, and provide meaningful analytical insights such as monthly patient visits, common diseases, and revenue analysis. The system uses SQL for data definition and manipulation and applies normalization principles to maintain consistency and eliminate anomalies.

This project aims to showcase the practical implementation of ER modeling, relational schema design, primary and foreign key constraints, and healthcare data analytics using DBMS concepts.

OVERVIEW

Introduction

Healthcare sector generates large volumes of data daily, including patient details, medical history, doctor information, and billing records. Managing this data manually can lead to errors, redundancy, and inefficiency. A database-driven healthcare system ensures structured storage, secure access, and efficient data retrieval.

Objectives

- To manage patient and doctor records
- To schedule and track appointments
- To store diagnosis and treatment details
- To maintain billing and payment information
- To generate healthcare analytics reports

Scope of this Healthcare Analytics System

The system is designed for hospitals and clinics to manage operational and analytical data efficiently using relational database concepts.

ENTITY RELATIONSHIP (ER) DIAGRAM

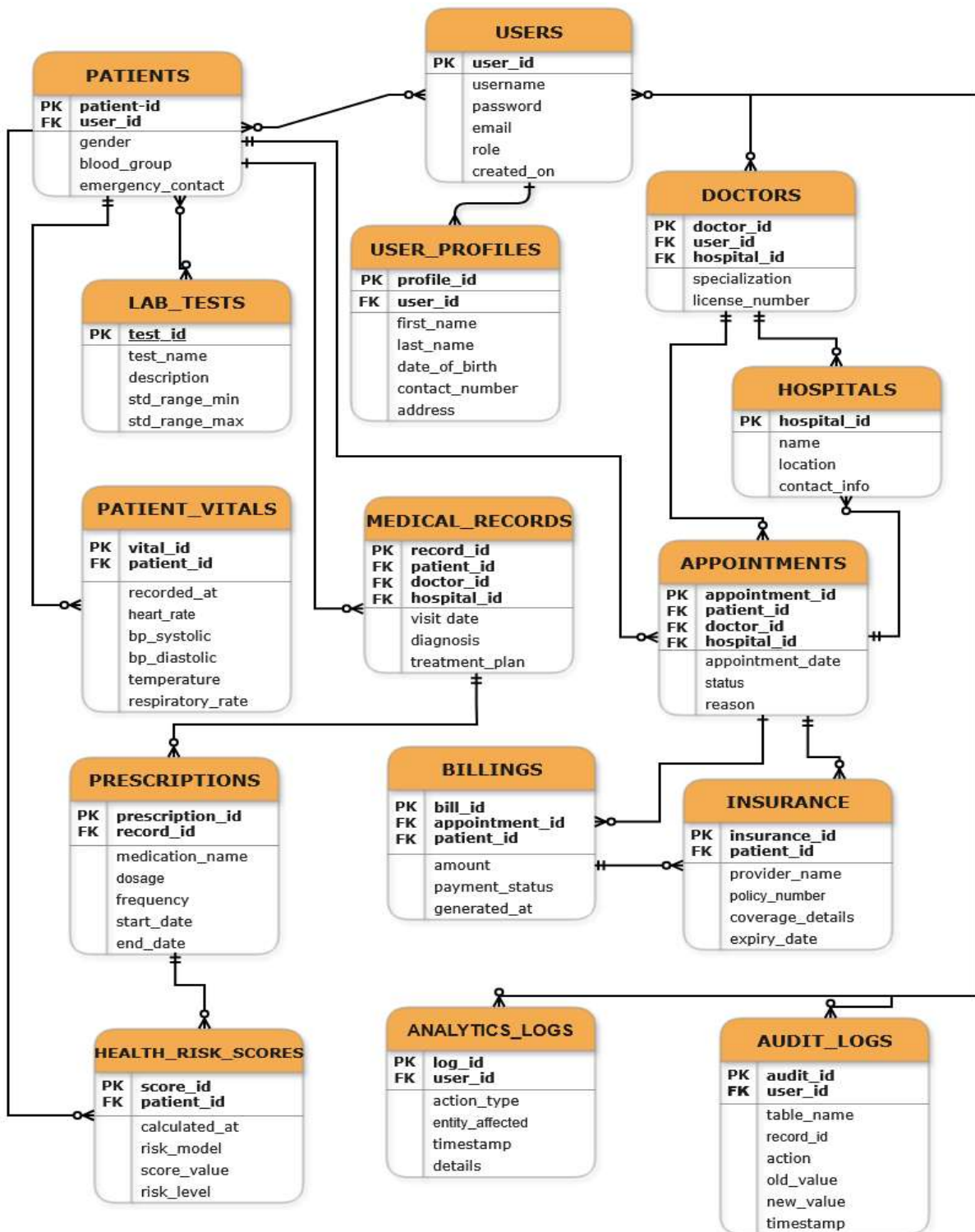


Figure 1: ER Diagram of Healthcare Analytics System

The ER diagram is represented using Crow's Foot notation to indicate cardinality relationships.

Description of ER Diagram (Figure 1)

The above ER diagram represents the structural design of the Healthcare Analytics System. The system consists of major entities such as Patients, Users, Doctors, Hospitals, Appointments, Medical_Records, Lab_Tests, Prescriptions, Billings, Insurance, and Analytics_Logs.

The Users entity manages authentication and role-based access control. Patients and Doctors are linked to Users through foreign key relationships. Appointments act as a connecting entity between patients, doctors, and hospitals.

Medical_Records store diagnosis and treatment-related information. Prescriptions are linked to medical records, and Billings are generated based on appointments and patient services. Insurance details are maintained for coverage tracking. Audit_Logs and Analytics_Logs are included to support system monitoring and analytical processing.

Primary keys uniquely identify each entity, and foreign keys maintain referential integrity across relationships.

RELATIONAL SCHEMA

PATIENTS

PATIENT_ID	PRIMARY KEY
USER_ID	FOREIGN KEY
GENDER	
BLOOD_GROUP	
EMERGENCY_CONTACT	

USERS

USER_ID	PRIMARY KEY
USERNAME	
PASSWORD	
EMAIL	
ROLE	
CREATED_ON	

USER PROFILES

PROFILE_ID	PRIMARY KEY
USER_ID	FOREIGN KEY
FIRST_NAME	
LAST_NAME	
DATE_OF_BIRTH	
CONTACT_NUMBER	
ADDRESS	

DOCTOR

ORDER_ID	PRIMARY KEY
USER_ID	FOREIGN KEY
HOSPITAL_ID	FOREIGN KEY
SPECILIZATION	
LICENSE_SYSTEM	

HOSPITALS

HOSPITAL_ID	PRIMARY KEY
NAME	
LOCATION	
CONTACT_INFO	

APPOINTMENTS

APPOINTMENT_ID	PRIMARY KEY
PAITENT_ID	FOREIGN KEY
DOCTOR_ID	FOREIGN KEY
HOSPITAL_ID	FOREIGN KEY
APPOINTMENT_DATE	
STATUS	
REASON	

MEDICAL RECORDS

RECORD_ID	PRIMARY KEY
PATIENT_ID	FOREIGN KEY
DOCTOR_ID	FOREIGN KEY
HOSPITAL_ID	FOREIGN KEY
VISIT_DATE	
DIAGNOSIS	
TREATMENT_PLAN	

LAB TESTS

TEST_ID	PRIMARY KEY
TEST_NAME	
DESCRIPTION	
STD_RANGE_MIN	
STD_RANGE_MAX	

PATIENT VITALS

VITAL_ID	PRIMARY KEY
PATIENT_ID	FOREIGN KEY
RECORDED_AT	
HEART_RATE	
BP_SYSTOLIC	
BP_DIASTOLIC	
TEMPERATURE	
RESPIRATORY_RATE	

PRESCRIPTIONS

PRESCRIPTION_ID	PRIMARY KEY
RECORD_ID	FOREIGN KEY
MEDICATION_NAME	
DOSAGE	
FREQUENCY	
START_DATE	
END_DATE	

BILLINGS

BILL_ID	PRIMARY KEY
APPOINMENT_ID	FOREIGN KEY
PATIENT_ID	FOREIGN KEY
AMOUNT	
PAYMENT_STATUS	
GENERATED_AT	

INSURANCE

INSURANCE_ID	PRIMARY KEY
PATIENT_ID	FOREIGN KEY
PROVIDER_NAME	
POLICY_NUMBER	
COVERAGE_DETAILS	
EXPIRY_DATE	

HEALTH RISK SCORES

SCORE_ID	PRIMARY KEY
PATIENT_ID	FOREIGN KEY
CALCULATED_AT	
RISK_MODEL	
SCORE_VALUE	
RISK_LEVEL	

ALALYTICS LOGS

LOG_ID	PRIMARY KEY
USER_ID	FOREIGN KEY
ACTION_TYPE	
ENTITY_AFFECTED	
TIMESTAMP	
DETAILES	

AUDIT LOGS

AUDIT_ID	PRIMARY KEY
USER_ID	FOREIGN KEY
TABLE_NAME	
RECORD_ID	
ACTION	
OLD_VALUE	
NEW_VALUE	
TIMESTAMP	